AC – DC CONVERTERS

RECTIFIERS

A rectifier converts ac power to dc power, which is known as rectification.

Uncontrolled Rectifier / Controlled rectifier

i. Single-phase half-wave rectifiers
ii. Single-phase full-wave rectifiers
iii. Three-phase rectifiers

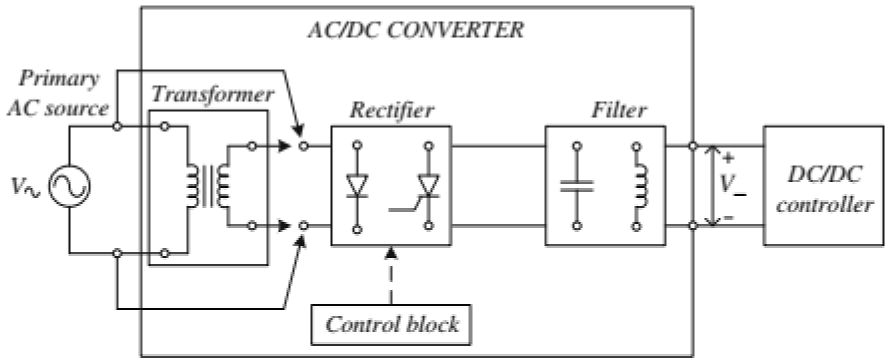
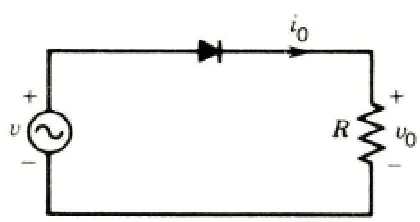


Figure 3: Block diagram of AC/DC Converter

Diode Rectifier

Single phase rectifiers with resistive load



(a)

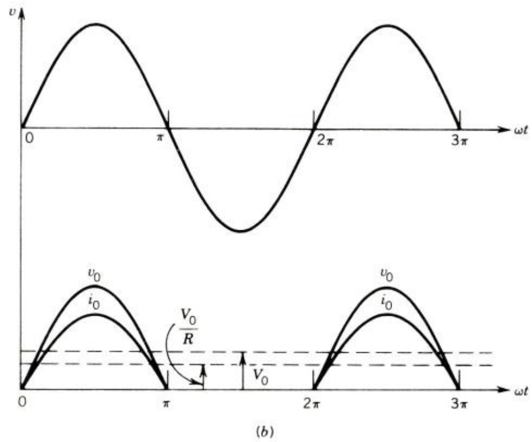


Figure 4: Single-phase diode rectifier with R load. (a) circuit (b) waveform

Instantaneous phase voltage and current is given as

$$v(t) = \sqrt{2}V_{rms}\sin(\omega t)$$

$$i(t) = \sqrt{2}i_{rms}\sin(\omega t)$$

The average value of the voltage which represents the dc component of the signal is

$$V_0 = \frac{1}{2\pi}\int_0^{\pi} V_m \sin(\omega t)dt$$

$$V_0 = \frac{V_m}{\pi}$$

$$i_0 = \frac{V_0}{R}$$

Note that $V_m = \sqrt{2}V_{rms}$

The rms values of voltage and current are

$$V_{rms} = \left[\frac{1}{2\pi}\int_0^{\pi} V_m^2 \sin^2(\omega t)dt\right]^{0.5} = \frac{V_m}{\sqrt{2}}$$

Resistive and Inductive Load

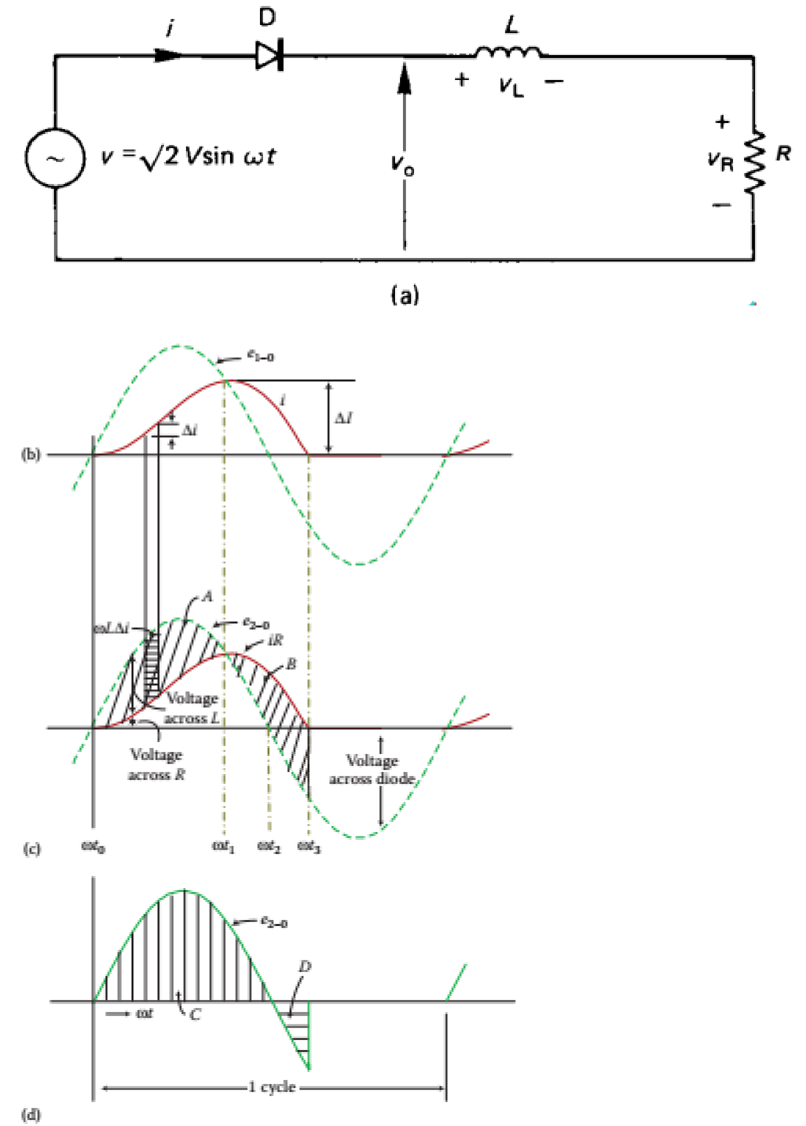


Figure 5: Single-phase half-wave diode rectifier with R - L load: (a) circuit (b) input voltage (c) input current (d) output voltage.

It can be seen that the load current flows not only in the positive half-cycle of the supply voltage but also in a portion of the negative half-cycle of the supply voltage. The load inductor SE maintains the load current and the inductor's terminal voltage so as to overcome the negative supply and keep the diode forward biased and conducting. Area A is equal to area B if figure c. during diode conduction the following equations are available.

$$\sqrt{2}V\sin\omega t = Ri + L\frac{di}{dt}$$

$$\frac{\sqrt{2}V}{L}\sin\omega t = \frac{R}{L}i + \frac{di}{dt}$$

Single-phase diode full-wave rectifier

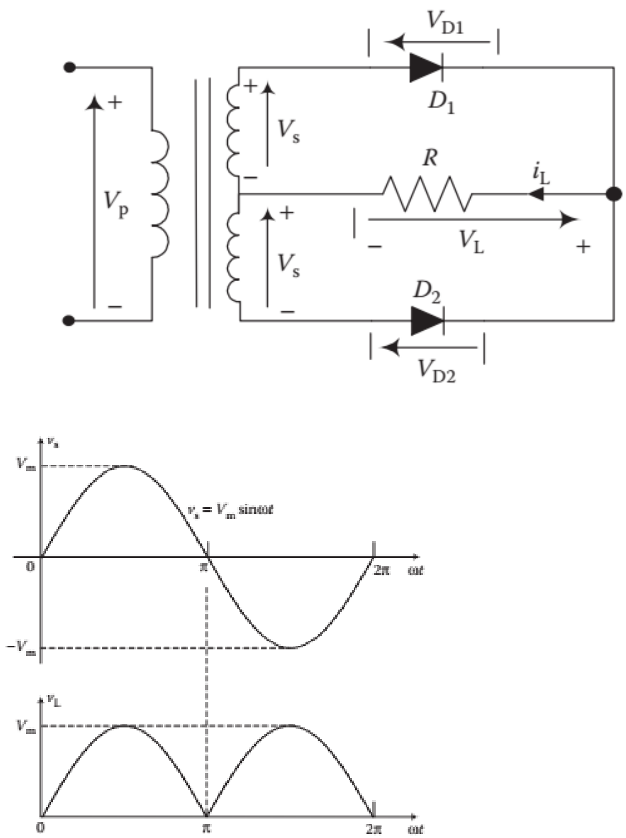
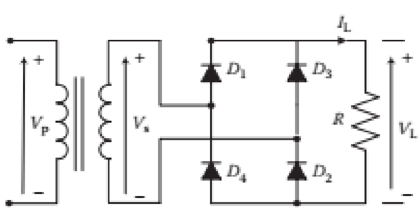


Figure 6: Center-tap rectifier (a) circuit (b) waveforms

$$V_0 = \frac{1}{\pi}\int_0^{\pi} V_m \sin(\omega t)dt = \frac{2V_m}{\pi}$$

$$i_0 = \frac{V_0}{R}$$

R – Load



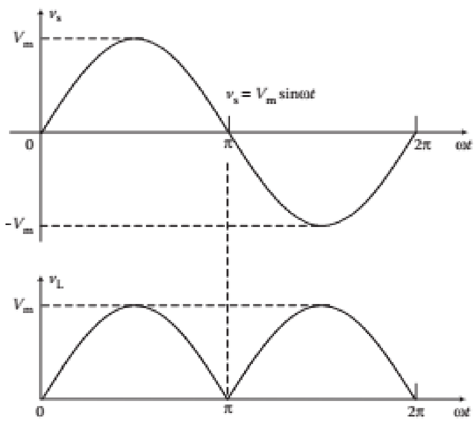


Figure 7: Bridge rectifier (a) circuit (b) waveforms

The output average voltage is

$$V_0 = \frac{1}{\pi}\int_0^{\pi}\sqrt{2}V\sin\omega t(\omega t) = \frac{2\sqrt{2}V}{\pi} = \frac{2V_m}{\pi}$$

$$i_0 = \frac{V_0}{R}$$

$$V_{rms} = \left[\frac{1}{2\pi}\int\left(2\sqrt{2}\sin(\omega t)\right)^2 d(\omega t)\right]^{0.5} = V\left[\frac{1}{\pi}\int_0^{\pi}\sin\theta^2 d\theta\right]^{0.5} = V$$

$$i_{rms} = \frac{V_{rms}}{R} = \frac{V}{R}$$

Phase Controlled Rectifiers

In phase controlled rectifiers thyristors are used instead of diodes. The average value of the output voltage is controlled by controlling the start of thyristor conduction. This will be considered in the example of a half-wave thyristor rectifier

Single-Phase half-Wave controlled rectifier with R load

Since, the thyristor conducts only for a positive half cycle, it's called a half-wave rectifier.

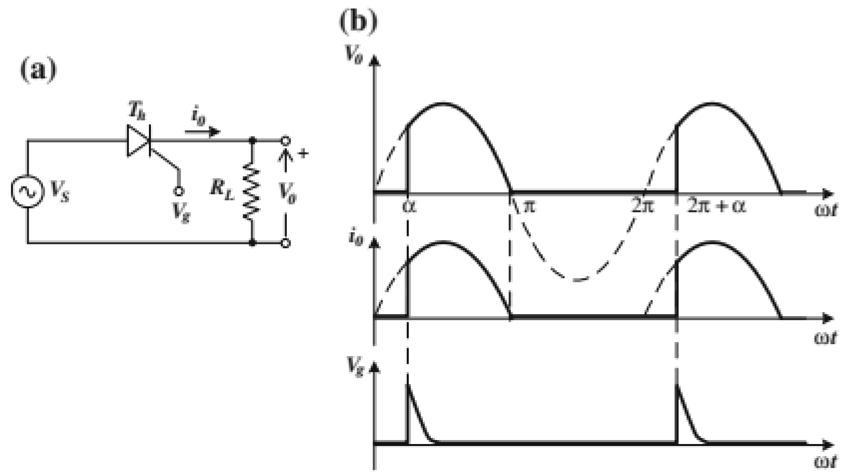


Figure 8: A half-wave thyristor rectifier (a) and the voltage and current waveforms (b)

Working: During the positive half cycle of input AC supply vs (=Vm sin ωt), the thyristor T is in forward blocking mode (FBM), whereas during the negative half cycles of supply it is in reverse blocking mode (RBM).

Thyristor starts conducting when a positive gate pulse is applied to it during the FBM. Thyristor rectifier with resistive load A simple thyristor rectifier circuit consisting of a single thyristor and a resistive load is shown below.

Therefore, let at ωt = α, thyristor T is triggered, and it conducts up to π. So, the load gets connected with the source, and load current i0 in phase with the load voltage v0 flows through Vs, T, and the load. At ωt = π, as the supply voltage becomes zero, the load current also becomes zero, and therefore thyristor T gets turned off at ωt = π due to the natural reversal of supply voltage, that is, the natural or line commutation, as shown in

Thyristor T does not conduct from π to 2π since it is reverse biased. Again, at ωt = 2π + α, when the thyristor is triggered, it starts conducting, and in this way the cycle repeats. Therefore, by controlling the firing angle, the average output voltage can be controlled that is, as the firing angle increases, the average load voltage decreases.

A single-phase HWR is one that produces only one pulse of load voltage or current during one cycle of source voltage, (Figure …b). So, it is also called the single-phase one-pulse rectifier. The variation of voltage across the thyristor is also shown as vT in Figure ….b. When the thyristor conducts, voltage across it (vT) is ideally zero, (practically vT = 1 to 1.5 V).

But during the off period of the thyristor, voltage across the thyristor (vT) has the wave shape of supply voltage vs. Therefore, it can be seen that vs = v0 + vT

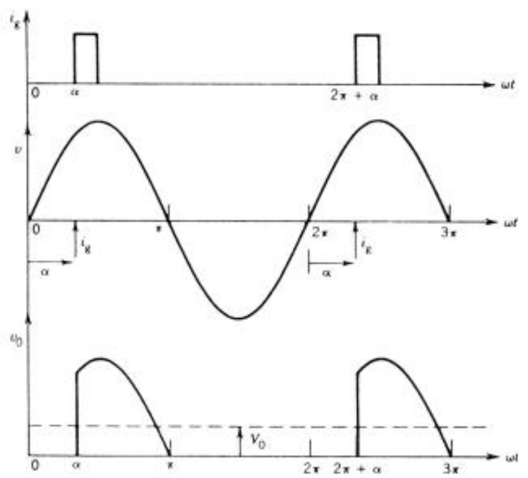
The average output voltage

$$V_0 = \frac{1}{2\pi}\int_{\alpha}^{\pi}\sqrt{2}V\sin(\omega t)dt = \frac{\sqrt{2}V}{2\pi}\left(1+\cos\alpha\right) = 0.45V\,\frac{1+\cos\alpha}{2}$$

The output rms voltage is

$$V_{rms} = \left[\frac{1}{2\pi}\int_{\alpha}^{\pi}(\sqrt{2}V\sin(\omega t))^2 d(\omega t)\right]^{0.5} = V\left[\frac{1}{\pi}\int_{\alpha}^{\pi}(\sin\theta)d\theta\right]^{0.5} = V\left[\frac{1}{\pi}\left(\frac{\pi-\alpha}{2}+\frac{\sin 2\alpha}{4}\right)\right]^{0.5}$$

$$i_{rms} = \frac{V_{rms}}{R}$$



Thyristor rectifier with inductive load

Most practical loads have resistance R and inductance L. for example, the armature of a dc motor load has resistance and inductance. The field circuit of a dc motor is highly inductive. A thyristor rectifier circuit with a load consisting R and L is shown below.

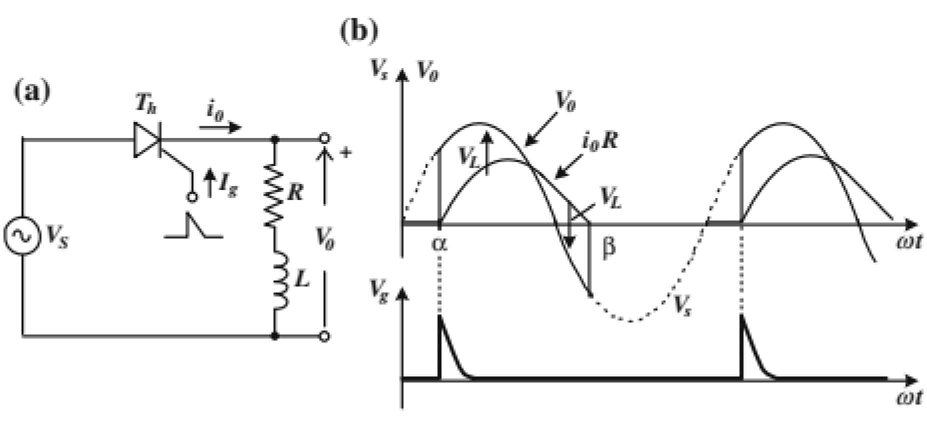
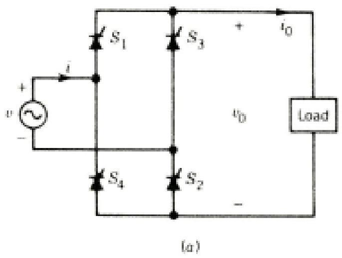


Figure 9. Thyristor rectifier with inductive load. (a) circuit (b) waveform

Thyristor full-wave rectifier center-tapped transformer

$$V_0 = \frac{1}{\pi}\int_{\alpha}^{\pi} \sqrt{2}V \sin(\omega t) d(\omega t) = \frac{\sqrt{2}V}{\pi}(1+\cos\alpha)$$

Thyristor full-wave rectifier



$$V_0 = \frac{1}{\pi}\int_{\alpha}^{\alpha+\pi} V_m \sin\omega t . d(\omega t) = \frac{2V_m}{\pi}\cos\alpha$$

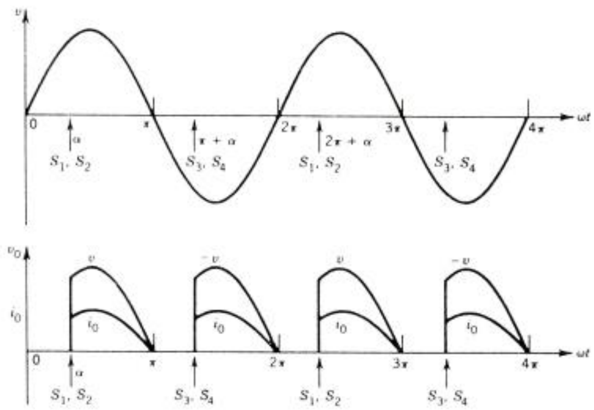
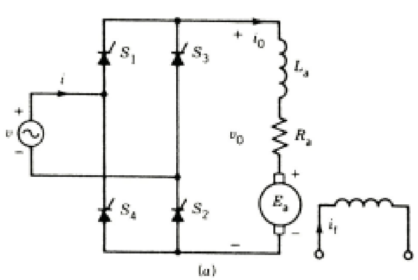


Figure () Thyristor full-wave rectifier with R load. (a) circuit (b) waveform



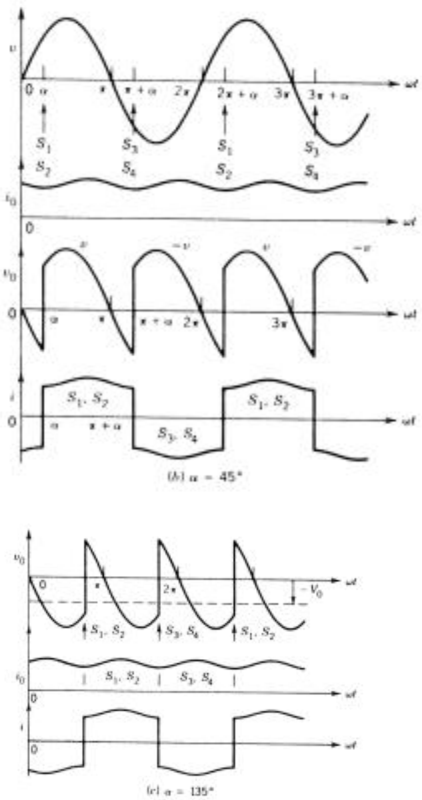
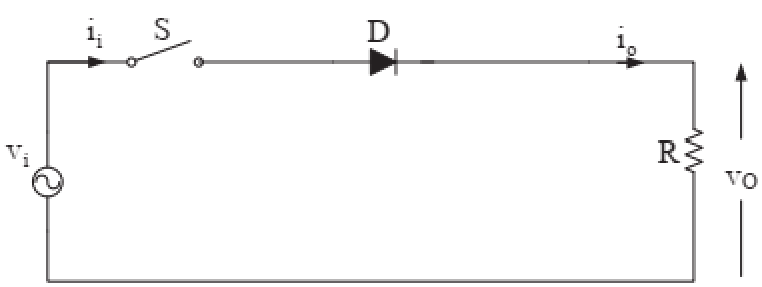


Figure 10: Thyristor full-wave rectifier with dc motor. (a) circuit (b) waveform with 45^0 and 135^0 firing angle.

Example

For the circuit shown below, $v_i = 120\sqrt{2}\sin\omega t$, and $R = 5\Omega$. When the switch S is closed, calculate the following:

Output voltage, output current, average output voltage, average output current, power and power factor.



SOLUTION

Output voltage and current

$$\overline{V}_o = \frac{1}{2\pi}\int_0^{\pi} \sqrt{2}\widetilde{V}_i \sin(\omega t)d(\omega t) = \frac{\sqrt{2}\widetilde{V}_i}{\pi} = \frac{120\sqrt{2}}{\pi} = 54 \text{ V}$$

$$\overline{I}_o = \frac{\overline{V}_o}{|Z_{o,0}|} = \frac{\overline{V}_o}{\sqrt{R^2 + (0 \times L)^2}} = \frac{\overline{V}_o}{R} = \frac{54}{5} = 10.8 \text{ A}$$

Average output voltage and current

$$\widetilde{V}_o = \left[\frac{1}{2\pi}\int_0^{\pi} \left(\sqrt{2}\,\widetilde{V}_i \sin(\omega t)\right)^2 d(\omega t)\right]^{1/2} = \frac{\sqrt{2}\widetilde{V}_i}{2} = \frac{120\sqrt{2}}{2} = 84.9 \text{ V}$$

$$\widetilde{I}_o = \widetilde{I}_i = \frac{\widetilde{V}_o}{R} = \frac{84.85}{5} = 17 \text{ A}$$

Assuming that the circuit is ideal, the input active power is equal to the power consumed by the load resistor and is given by:

$$P_i = P_o = \frac{\widetilde{V}_o^2}{R} = \widetilde{I}_o^2 R = (17)^2(5) = 1445 \text{ W(ideal operation)}$$

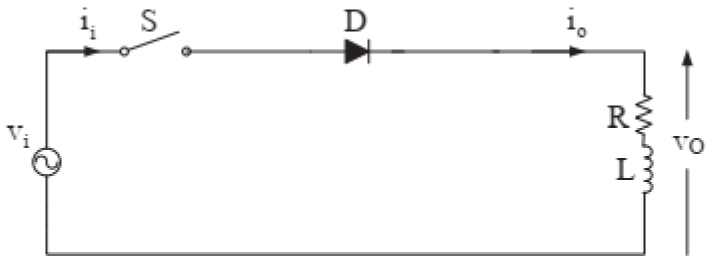
Therefore, the power factor at the ac side is:

$$\lambda = \frac{P_i}{S_i} = \frac{P_o}{\widetilde{V}_i \widetilde{I}_i} = \frac{1445}{(120)(17)} = 0.708$$

Example

For the circuit shown below, $v_i = 100\sin 377t$, $L = 0.1H$ and. $R = 100\Omega$ When the switch S is closed, perform the following:

(a) Determine the current function
(b) Calculate the average current
(c) Calculate the rms current
(d) Calculate the power consumed by the load
(e) Calculate the power factor



SOLUTION

For the given circuit the following results are obtained:

$$|Z_o| = \sqrt{R^2 + (\omega L)^2} = \sqrt{(100)^2 + (377 \times 0.1)^2} = 106.9\ \Omega$$

$$\varphi = \tan^{-1}\left(\frac{\omega L}{R}\right) = \tan^{-1}\left(\frac{377 \times 0.1}{100}\right) = 20.7^\circ \text{ and } \tau = \frac{L}{R} = \frac{0.1}{100} = 1 \times 10^{-3}\ \text{s.}$$

a) Substituting the previous results into

$$i_o = i_i = \frac{\sqrt{2}\tilde{V}_i}{|Z_o|}\left[\sin(\omega t - \varphi) + e^{-(R/L)t}\sin\varphi\right] \quad \text{for } 0 < \omega t < \pi$$

the current function is given by:

$$i(\omega t) = \frac{100}{106.9}\left[\sin(\omega t - 20.7^\circ) + (\sin 20.7^\circ)e^{-\frac{\omega t}{377 \times 0.001}}\right] \quad \text{for } 0 \le \omega t \le \beta$$

or

$$i(\omega t) = 0.935 \sin(\omega t - 20.7^\circ) + 0.353e^{-\frac{\omega t}{0.377}}$$

Using the above equation and the initial condition i(b) ¼ 0, the value of b can be calculated:

$$\sin(\beta - 20.7°) + (\sin(20.7°))e^{-\frac{\beta}{0.3777}} = 0$$

which is found to be $\beta = 201°$.

b) $\bar{I} = \frac{1}{2\pi}\int_{0°}^{201°}\left[0.935\sin(\omega t - 20.7°) + 0.331e^{-\frac{\omega t}{0.377}}\right]d\omega t = 0.308\ A$

c) $\tilde{I} = \left[\frac{1}{2\pi}\int_{0°}^{201°}\left[0.935\sin(\omega t - 20.7°) + 0.331e^{-\frac{\omega t}{0.377}}\right]^2 d\omega t\right]^{1/2} = 0.474\ A$

d) Assuming that the circuit operates without losses (i.e., is ideal), then

input active power ¼ output active power ¼ Po

$$\begin{aligned} P_o &= \frac{1}{2\pi}\int_0^{2\pi} v_o(\omega t)\cdot i(\omega t)d(\omega t) \\ &= \frac{1}{2\pi}\int_{0°}^{201°}[100\sin(\omega t)]\left[0.935\sin(\omega t - 20.7°) + 0.331e^{-\frac{\omega t}{0.377}}\right]d(\omega t) = 22.4\ W \end{aligned}$$

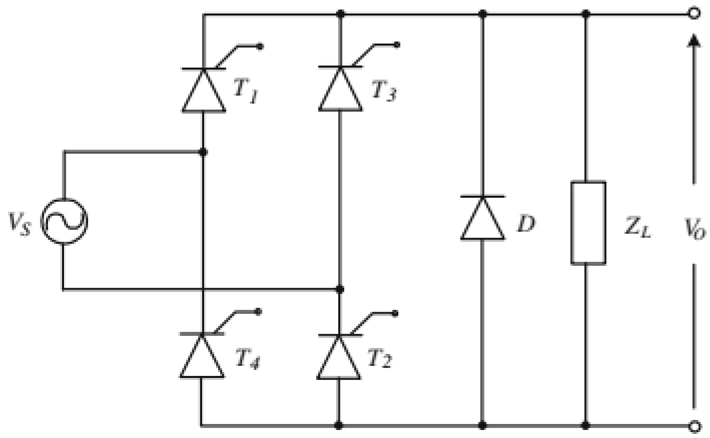
The same result can be obtained using the following equation:

$$P_o = \tilde{I}^2 R = (0.474)^2(100) = 22.4\ W$$

e) $\lambda = \frac{P_i}{S_i} = \frac{P_o}{\tilde{V}_i\tilde{I}} = \frac{22.4}{\left(\frac{100}{\sqrt{2}}\right)(0.474)} = 0.67$

Example

For the rectifier in Figure below determine the average value of the voltage across the load, the average load current, and the average values of the thyristor and the diode currents if the load is:
(a) resistance R,
(b) a series connection of the resistance R and the inductance L.
where VSM = 311 V, f = 50 Hz, R = 10 Ω, ωL > R, and α = π/3



Solution

(a) The load is the resistor R.
The average value of the voltage across the load is

$$V_o = \frac{1}{\pi}\int_{\alpha}^{\pi} V_m \sin(\omega t) d(\omega t)$$

$$V_o = \frac{V_m}{\pi}(1+\cos\alpha)$$

$$V_o = \frac{311}{\pi}(1+0.5) = 148.5V$$

The average value of the load current is

$$I_o = \frac{V_o}{R} = 14.85A$$

For a purely resistive load, there is no current through the diode, so $I_{do} = 0$
The average value of the thyristor current is one half the load current.

(b) The load is a series R-L circuit.
The average values of the load voltage and the load current are the same as in the case (a). Since the load is inductive, the current through the load is continuous and the diode D conducts when all thyristors in the bridge are off.
The average value of the current through the diode is

$$I_{do} = \frac{\alpha}{\pi} * I_o = 4.95A$$

and the average value of the current through the thyristor is also

$$I_{to} = \frac{(\pi - \alpha)}{\pi} * I_o = 4.95A$$

THREE-PHASE RECTIFIER

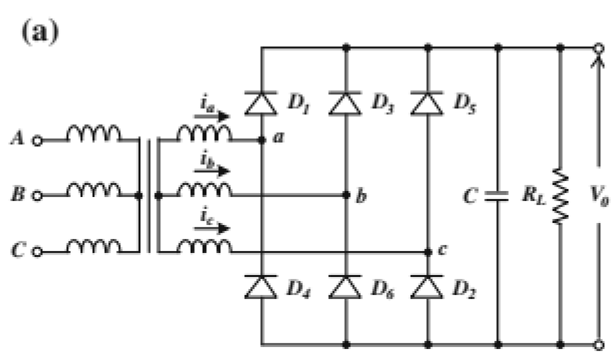
In industrial applications, where three-phase AC voltages are available, it is recommendable to use three-phase rectifiers. At the output of this type of rectifier compared to the single-phase rectifiers, the DC component is higher, the ripple of the output voltage is lower, and the output power is higher. Three-phase rectifiers have favorable features for equipment and installations requiring high power, where very high DC currents and relatively high voltages are necessary.

Hal-Wave Diode Rectifier

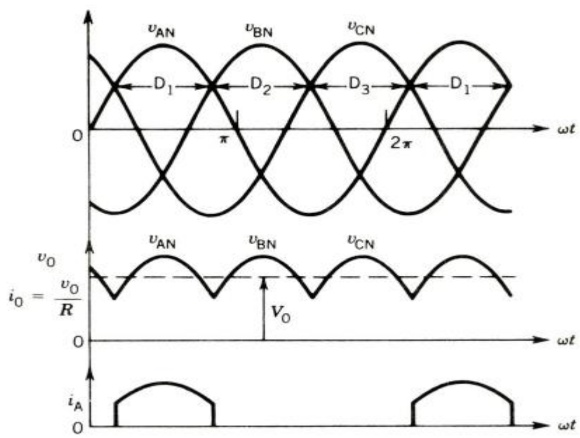
The voltages of the input three-phase-supply are:

$$\begin{cases} V_{AN} = V_m \sin \omega t \\ V_{BN} = V_m \sin(\omega t - 120^0) \\ V_{CN} = V_m \sin(\omega t + 120^0) \end{cases}$$

The diodes conduct in the sequence D1, D2,D3, D1, ... they conduct one at a time for 120 degree intervals. At any time the diode whose anode is at the highest instantaneous supply voltage will conduct. For insatnace, during the interval $30 < \omega t < 150$ Van is higher than both Vbn and Vcn. During this interval D1 conducts. When D1 conducts, the voltage across D2 is Vba and that across D3 is Vca. Diode D2 and D3 remain reverse-biased. When D1 conducts, the output load voltage V0 is the sa me as the phase voltage Van.



(a)



(b)

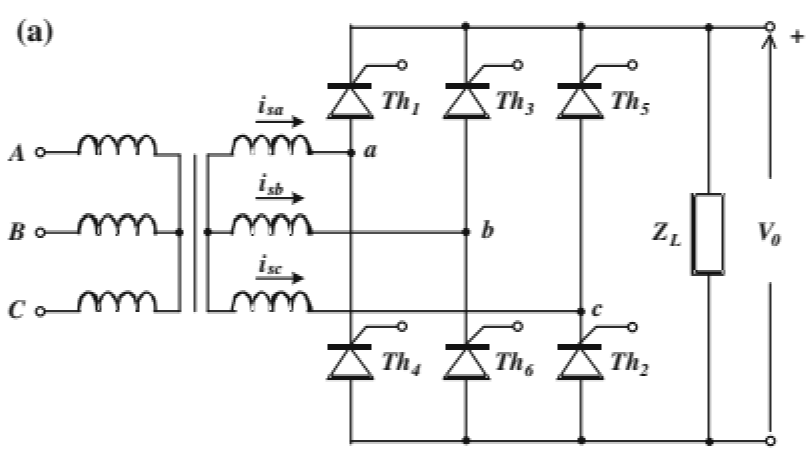
Figure (): Three-phase half-wave rectifier circuit with R-Load (a) circuit (b) Waveforms

The average value of the output voltage is

$$V_o = \frac{1}{\pi/3} \int_{\pi/3}^{2\pi/3} V_{LL} \sin(\omega t)\, d(\omega t) = \frac{3V_{LL}}{\pi}$$

Half-wave Thyristor-controlled Rectifier

The topology of the thyristor bridge rectifier is the same as that of the corresponding diode rectifier. The difference in the principle of operation is that the diodes are conducting when forward biased and the thyristors conducts if a triggering pulse is present in the gate circuit together with the forward bias.



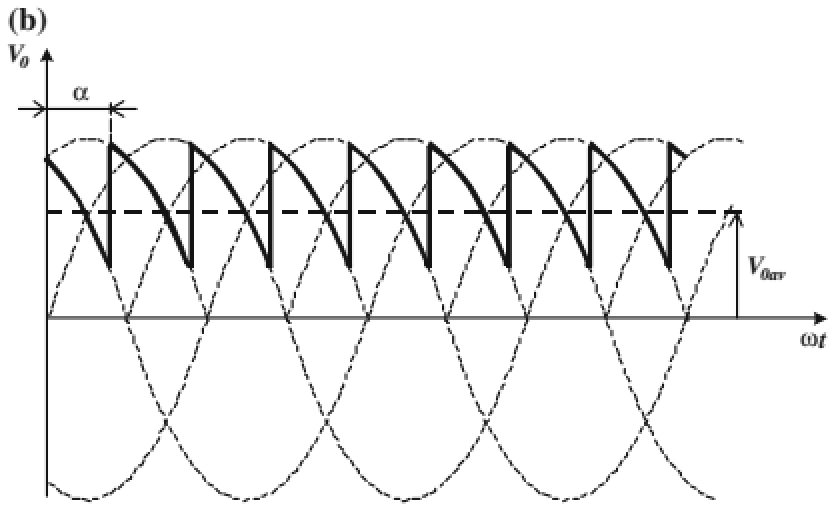


Figure (): A three-phase thyristor bridge rectifier (a) and the output voltage waveforms for a resistive load (b)

This is the way of controlling the angle of conduction and consequently the average value of the output voltage. If the angle of triggering is α, the output voltage waveform is as shown in Figure ()b above and its average value is determined by

$$V_o = \frac{1}{\pi/3}\int_{\pi/3+\alpha}^{2\pi/3+\alpha} V_{LL}\sin(\omega t)\,d(\omega t) = \frac{3V_{LL}}{\pi}\cos\alpha$$